The next matching level for triangle-free graphs with bounded degree and an exact Bougard–Joret value

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github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

Let $T(d, m)$ be the maximum number of edges in a triangle-free graph with maximum degree at most d and matching number at most m, with no restriction on its order. We prove $T(d, d+1) = d^2 + d + 2$ for every integer $d \geq 7$. A shortest-odd-cycle deficit bound, followed by a five-type equality obstruction, first gives the same maximum on $2d + 3$ vertices. The Tutte–Berge formula extends this bound to arbitrary order. The attaining construction is due to Banak, Ekim and Taşkın; the contribution is the uniform upper bound. This settles the next-matching slice of the Ahanjideh–Ekim–Yıldız conjecture and, in particular, resolves the published bounds $184 \leq T(13, 14) \leq 185$ in favor of 184. As a consequence, the Bougard–Joret extremal function satisfies $f(2d + 3, 2, d + 2) = d^2 + 4d + 2$ for every $d \geq 7$, an unbounded strict-interior discrepancy from the degree-sum prediction. The lower value is excluded by a structural theorem: every triangle-free graph on $2d + 3$ vertices with maximum degree at most d and $d^2 + d + 2$ edges has two vertices with the same d neighbours, so its complement is not $(d + 2)$-connected. The full AEY conjecture remains open. Exact finite certificates and independent implementation checks support, but do not replace, the proof.

1 Problem and relation to prior work

All graphs are finite, simple and undirected. Write $\mathrm{e}(F)$, $\Delta(F)$ and $\nu(F)$ for the number of edges, maximum degree and maximum matching size of F. Define

$$T(d, m) = \max\{\mathrm{e}(F) : F \text{ triangle-free}, \ \Delta(F) \leq d, \ \nu(F) \leq m\}.$$

Isolated vertices are allowed and have no effect on the maximum. We use T to distinguish this function from the Bougard–Joret minimum-edge function in Section 5.

Ahanjideh, Ekim and Yıldız (AEY) study this degree/matching problem and conjecture, in their intermediate range $d < i < Z(d)$, that $T(d,i) = di + i - d + 1$ [1, Conjecture 6.1]. Here $Z(d)$ is their threshold for a triangle-free factor-critical graph with their extremal degree sequence. Our result concerns the complete next level $i = d + 1$ for $d \geq 7$.

Banak, Ekim and Taşkın (BET) provide the attaining family in Proposition 4.1 and computationally resolve a substantial finite range [2]. Their Table 3 already determines $T(d, d+1)$ for $d = 7, \ldots, 12$. At $(d, m) = (13, 14)$ it reports lower bound 184 and upper bound 185. Those finite results and the construction are prior work. The new conclusion here is the uniform upper bound, including

$$T(13, 14) = 184.$$

The source comparison used AEY's final published text and BET's complete author manuscript; BET's final publisher PDF was not obtained. Additional searches located no explicit prior uniform solution of this slice, without establishing exhaustive bibliographic priority.

Regular triangle-free extremal results, including Caro–Tuza [3, Theorem 2(iii)] and the later regular Turán literature [4], already imply strong regular-degree obstructions. They are not asserted as new here. Our fixed-order theorem permits nonregular graphs with maximum degree at most d; its attaining graph has one vertex of degree four and all other vertices of degree d.

Theorem 1 (Next matching level, D)**.** *For every integer $d \geq 7$,*

$$T(d, d+1) = d^2 + d + 2.$$

Our universal deductions are labeled [D] and exact finite checks [MV]. The proof and certificates are archived as computation EXP-003 [10]. The first interior shell of the Bougard–Joret problem is the subject of the companion paper [9].

2 The fixed-order upper bound

Lemma 2 (Shortest odd cycle)**.** *Let F be triangle-free on n vertices and let C be a shortest odd cycle, of length ℓ. Then $\ell \geq 5$, C is chordless, every vertex outside C has at most two neighbors on C, and*

$$\sum_{x \in V(C)} \deg_F(x) \leq 2n.$$

Proof. Triangle-freeness gives $\ell \geq 5$. A chord would split C into two cycles of opposite parity, producing a shorter odd cycle. If an outside vertex has at least three neighbors on C, list all of them in cyclic order. The intervening arc lengths sum to the odd number ℓ, so at least one arc is odd. Every arc has length at least two, by triangle-freeness. The other arcs have total length at least four, so the odd arc has length at most $\ell - 4$. Adding the two edges to the outside vertex produces an odd cycle of length at most $\ell - 2$, a contradiction. The cycle contributes 2ℓ to its degree sum and the $n - \ell$ outside vertices contribute at most $2(n - \ell)$, proving the bound. □

If $\Delta(F) \leq d$, its total degree deficit is

$$D(F) = \sum_{x \in V(F)} (d - \deg_F(x)) = nd - 2\mathrm{e}(F).$$

Every summand is nonnegative. For nonbipartite F, Lemma 2 yields

$$D(F) \geq \sum_{x \in V(C)} (d - \deg_F(x)) \geq \ell d - 2n \geq 5d - 2n. \tag{1}$$

Theorem 3 (Fixed order, D)**.** *For every integer $d \geq 7$, the maximum number of edges in a triangle-free graph on $2d+3$ vertices with maximum degree at most d is d^2+d+2.*

Proof of the upper bound. Let $n = 2d+3$. If F is bipartite, one part has at most $d+1$ vertices, so summing degrees on that part gives $\mathrm{e}(F) \leq d(d+1)$. Otherwise (1) gives $D(F) \geq d-6$ and therefore

$$\mathrm{e}(F) \leq d^2+d+3. \tag{2}$$

Suppose equality holds. Then every inequality in (1) is an equality. Thus a shortest odd cycle has length five, every outside vertex has degree exactly d, and every outside vertex has exactly two neighbors on the cycle.

Number the cycle $c_0, \ldots, c_4$, with indices modulo five. Its two neighbors at an outside vertex must form a nonadjacent pair, uniquely $\{c_{i-1}, c_{i+1}\}$ for some i. Let V_i contain c_i and all outside vertices with this pair, and put $n_i = |V_i|$. Then

$$n_i \geq 1, \qquad \sum_i n_i = 2d+3. \tag{3}$$

Each V_i is independent, and there are no edges between V_i and V_{i+2}: such vertices share a cycle neighbor. Every possible edge therefore joins consecutive types. We do not assume completeness between types.

By definition, c_i is adjacent to every member of $V_{i-1} \cup V_{i+1}$ and no others. Hence

$$b_i := n_{i-1} + n_{i+1} = \deg_F(c_i) \leq d. \tag{4}$$

If $n_i \geq 2$, an outside vertex in V_i has degree d and all its neighbors lie in the two adjacent types. Thus

$$n_i \geq 2 \quad \Longrightarrow \quad b_i = d. \tag{5}$$

These constraints are impossible for $d > 6$. If every $n_i \geq 2$, summing (5) gives $5d = 2\sum_i n_i = 4d+6$, so $d = 6$. Otherwise rotate indices so that $n_0 = 1$.

If $n_1, n_4 \geq 2$, equation (5) gives $n_2 = n_3 = d-1$. The total in (3) forces $n_1 + n_4 = 4$, so $n_1 = n_4 = 2$. But then $b_2 = n_1 + n_3 = d+1$, contradicting (4).

We may therefore reflect indices and assume $n_0 = n_1 = 1$. If $n_2, n_4 \geq 2$, equation (5) gives $n_3 = d-1$. Now (3) forces $n_2 + n_4 = d+2$, contrary to $b_3 \leq d$.

Otherwise there are three consecutive singleton types; after relabeling, $n_0 = n_1 = n_2 = 1$. Then $n_3 + n_4 = 2d$, while (4) at types 0 and 2 gives $n_4, n_3 \leq d-1$, another contradiction.

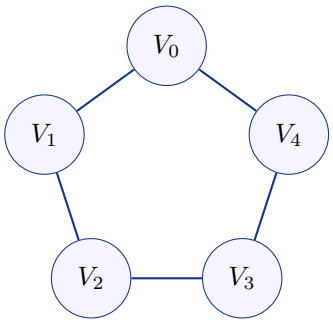
Equality in (2) is impossible. Integrality improves the upper bound to d^2+d+2. No parity assumption was used. Attainment is proved in Section 3. □

3 The known attaining construction

Put $p = d+1$. Start with the crown graph on classes $A = \{A_0, \ldots, A_{p-1}\}$ and $B = \{B_0, \ldots, B_{p-1}\}$: it contains A_iB_j exactly when $i \neq j$. Delete A_0B_1 and A_1B_0, then add a vertex v adjacent to A_0, A_1, B_0, B_1. Denote this graph by Q_d.

This is the $t = 1$ case of BET's construction [2, Proposition 4.1], written in crown notation. It is prior work. The four neighbors of v are independent, and $Q_d - v$ is bipartite, so Q_d is triangle-free. Every old vertex has degree d and v has degree four. Consequently, for $d \geq 7$,

$$|V(Q_d)| = 2d+3, \qquad \Delta(Q_d) = d, \qquad \mathrm{e}(Q_d) = d(d+1) - 2 + 4 = d^2+d+2. \tag{6}$$



$n_i = |V_i| \geq 1, \quad \sum_i n_i = 2d + 3$

$b_i = n_{i-1} + n_{i+1} \leq d$

$n_i \geq 2 \implies b_i = d$

These necessary equality constraints are incompatible when $d > 6$.

Figure 1: The five-type obstruction. Lines show allowed adjacencies, not an assumption of complete bipartite adjacency.

The edges $A_iB_{i+2 \bmod p}$ form a perfect matching on the old vertices. Since $p \geq 8$, they avoid the deleted diagonal and both further deleted edges. Thus $\nu(Q_d) \geq d + 1$, and its order gives $\nu(Q_d) \leq d + 1$. This proves attainment in Theorem 3 and the lower bound in Theorem 1.

Remark 4 (The degree-six boundary). The balanced blowup of C_5 with five parts of size three has order 15, degree six and 45 edges, exceeding $6^2 + 6 + 2 = 44$. Its matching number is at most seven by order. Thus neither theorem extends unchanged to $d = 6$.

4 Arbitrary order through Tutte–Berge

We first need the adjacent odd-order bound, also consistent with AEY's Theorem 3.4.

Lemma 5. *If J is triangle-free on $2d + 1$ vertices with $\Delta(J) \leq d$, then $\mathrm{e}(J) \leq d^2 + 1$.*

Proof. If J is bipartite, a smaller part has at most d vertices, so $\mathrm{e}(J) \leq d^2$. Otherwise (1) gives $D(J) \geq d - 2$ and hence $2\mathrm{e}(J) \leq (2d + 1)d - (d - 2) = 2d^2 + 2$. □

The classical Tutte–Berge formula [5, 6] states

$$\nu(F) = \min_{S \subseteq V(F)} \frac{|V(F)| + |S| - o(F - S)}{2}, \tag{7}$$

where $o(F - S)$ is the number of odd-order components of $F - S$. Choose a set S attaining this minimum. Writing $t_C = \lfloor |V(C)|/2 \rfloor$ for each component C of $F - S$ gives the identity

$$\nu(F) = |S| + \sum_C t_C. \tag{8}$$

Proof of Theorem 1. Let F be triangle-free, with $\Delta(F) \leq d$ and $\nu(F) \leq d + 1$, and take S as in (8). The number of edges incident to S is at most $\sum_{x \in S} \deg_F(x) \leq d|S|$. Edges inside S are counted twice in the degree sum, which is harmless for this upper bound.

Every remaining edge belongs to one of the components C. If C has even order $2t_C$, its degree bound gives $\mathrm{e}(C) \leq dt_C$. For odd order $2t_C + 1$, the bounds are

$$\mathrm{e}(C) \leq \begin{cases} t_C(t_C + 1) \leq dt_C, & t_C \leq d - 1, \\ dt_C + 1, & t_C = d, \\ dt_C + 2, & t_C = d + 1. \end{cases}$$

The first line is Mantel's classical triangle-free bound; the second is Lemma 5; the third is Theorem 3. Formula (8) ensures that $t_C \le d+1$ for every component and that at most one component has $t_C \ge d$, since $2d > d+1$. The total surplus over $d\sum_C t_C$ is therefore at most two. Hence

$$\mathrm{e}(F) \le d|S| + d\sum_C t_C + 2 = d\nu(F) + 2 \le d(d+1) + 2.$$

The known graph Q_d attains this value by (6). □

Remark 6 (A second reduction). One may instead discard isolated vertices and select an edge-extremizer with the largest number of d-star components. AEY Corollary 3.5 makes every other component factor-critical, with matching size $t \ge d$ and order $2t+1$. At budget $d+1$ there is at most one such component. If its matching size is d, Lemma 5 plus at most one additional star gives at most d^2+d+1 edges; if it is $d+1$, Theorem 3 gives d^2+d+2. With only stars the bound is $d(d+1)$. This corroborates the upper bound. The direct Tutte–Berge proof does not rely on this extremal structural reduction.

5 A Bougard–Joret consequence

Let $f(n,a,k)$ be the minimum number of edges in a k-connected graph of order n and independence number exactly a, in the notation of Bougard and Joret [7]. Das and Gupta disprove the original conjectured formula on the boundary $n = a+k$ [8]. The following consequence concerns strict-interior parameters. Corollary 7 first brackets the value within one edge; Theorem 8 then excludes the lower endpoint.

Corollary 7 (One-edge bracket, D). *For every integer $d \ge 7$,*

$$d^2+4d+1 \le f(2d+3, 2, d+2) \le d^2+4d+2. \tag{9}$$

Proof. If G has these parameters, then $F = \overline{G}$ is triangle-free. Since $\delta(G) \ge d+2$, it has $\Delta(F) \le d$. Theorem 3 gives

$$\mathrm{e}(G) = \binom{2d+3}{2} - \mathrm{e}(F) \ge d^2+4d+1.$$

For the upper bound, start with the crown $K_{d+1,d+1}$ minus its diagonal matching. Subdivide one existing edge ab by a new vertex w, and call the resulting graph P_d. Its old vertices have degree d and w has degree two. It is triangle-free: $P_d - w$ is bipartite and a, b are nonadjacent. It has an edge, so $G_d = \overline{P_d}$ has independence number exactly two. Moreover,

$$\mathrm{e}(P_d) = d(d+1)+1, \qquad \mathrm{e}(G_d) = d^2+4d+2.$$

It remains to prove $(d+2)$-connectivity of G_d. A smaller vertex cut would leave at least $d+2$ vertices partitioned into two nonempty sets with no G_d edges between them. Thus P_d would contain a complete bipartite subgraph on at least $d+2$ vertices.

If that subgraph excludes w, its sides lie in opposite original crown classes. The sets of matched indices used by the two sides are disjoint because diagonal edges are missing. Its total order is therefore at most $d+1$. If it includes w, its opposite side is contained in $\{a,b\}$. An opposite side of size one gives a star of order at most $\Delta(P_d)+1 = d+1$. If the opposite side is $\{a,b\}$, no old vertex can join w in its own side: an old vertex cannot be adjacent to both original classes in the bipartite graph $P_d - w$. This subgraph has order only three. All cases contradict $d+2$ surviving vertices. Therefore G_d is $(d+2)$-connected; its minimum degree also shows that its connectivity is exactly $d+2$. □

These parameters satisfy $2d+3 > 2+(d+2)$ and $2d+3 < 2(d+2)$, placing them strictly inside the originally proposed first regime. The excess of the lower endpoint over its degree-sum prediction is

$$(d^2+4d+1) - \left\lceil \frac{(2d+3)(d+2)}{2} \right\rceil = \left\lfloor \frac{d}{2} \right\rfloor - 2.$$

This is positive for $d \geq 7$ and unbounded. The next subsection shows that the upper endpoint in (9) is the value, so the excess is $\lfloor d/2 \rfloor - 1$; for example $f(19,2,10) = 98$, whereas the degree-sum prediction is 95.

5.1 The exact value

Theorem 8 (Exact value, D)**.** *For every integer $d \geq 7$,*

$$f(2d+3, 2, d+2) = d^2+4d+2.$$

By Corollary 7 and complementation, it suffices to prove the following structural statement. A complement $\overline{F}$ on $2d+3$ vertices fails to be $(d+2)$-connected exactly when F contains a complete bipartite subgraph $K_{U,W}$, with U, W nonempty and every U–W pair an edge, on at least $d+2$ vertices: deleting the other at most $d+1$ vertices leaves U and W with no edge of $\overline{F}$ between them.

Theorem 9 (Shared neighbourhood, D)**.** *Let $d \geq 7$ and let F be triangle-free on $n = 2d+3$ vertices with $\Delta(F) \leq d$ and $\mathrm{e}(F) = d^2+d+2$. Then two vertices of F have the same d neighbours; in particular $F \supseteq K_{2,d}$.*

Proof of Theorem 8 from Theorem 9. A graph G of order $2d+3$ with independence number two, connectivity at least $d+2$ and d^2+4d+1 edges would have a triangle-free complement F with $\Delta(F) \leq d$ and $\mathrm{e}(F) = d^2+d+2$. Theorem 9 gives a $K_{2,d}$ in F, on $d+2$ vertices, hence a vertex cut of G of size $d+1$. Fewer edges are excluded by Corollary 7, whose construction G_d attains d^2+4d+2. □

Structure. Let F be as in Theorem 9. Its deficit is $D(F) = nd - 2\mathrm{e}(F) = d-4$. It is not bipartite, since a bipartite F has at most $d(d+1)$ edges. A shortest odd cycle has length five: length at least seven would give $D(F) \geq 7d - 2n = 3d-6 > d-4$ by (1). Write it $c_0 \cdots c_4$, indices modulo five. For an outside vertex x let $\sigma(x) = 2 - |N(x) \cap C| \geq 0$, and put $a = \sum_{x \notin C} \sigma(x)$ and $b = \sum_{x \notin C} (d - \deg x)$. Since the cycle is chordless, $\sum_{v \in C} \deg v = 10 + 2(n-5) - a$, and therefore

$$\sum_{v \in C} (d - \deg v) = d - 6 + a, \qquad D(F) = d - 6 + a + b, \qquad a + b = 2. \tag{10}$$

As in the proof of Theorem 3, outside vertices with two cycle neighbours fall into types: V_i consists of c_i and the outside vertices adjacent to c_{i-1}, c_{i+1}; put $n_i = |V_i|$. Each V_i is independent and there is no edge between V_i and $V_{i\pm2}$. The remaining outside vertices form the set X of *defective* vertices, those with at most one cycle neighbour; each contributes at least one to a, so $r := |X| \leq 2$ and $\sum_i n_i = 2d+3-r$. Put $W_i = V_{i-1} \cup V_{i+1}$, $b_i = |W_i|$ and $\varepsilon_i = |N(c_i) \cap X|$. The vertex c_i is adjacent to all of W_i and to no other typed vertex, so

$$\deg c_i = b_i + \varepsilon_i \leq d. \tag{11}$$

A defective vertex adjacent to c_j has no neighbour in $V_{j\pm1}$, by triangle-freeness.

Lemma 10 (Helpers, D)**.** *Let $v \in V_i$ be an outside vertex of degree d. If every neighbour of v in X is adjacent to c_i, then $N(v) \subseteq N(c_i)$, and $\{c_i, v\}$ together with $N(v)$ span a $K_{2,d}$.*

Proof. The typed neighbours of v lie in consecutive types, so $N(v) \subseteq W_i \cup X$; both parts lie in $N(c_i)$, the second by hypothesis, and $c_i \notin N(v)$ because V_i is independent. □

Call $x \in X$ a *helper* of an outside $v \in V_i$ if $x \sim v$ and $x \not\sim c_i$. Assume, for contradiction, that no two vertices of F share d neighbours. Then every outside typed vertex of degree d has a helper. There are $2d - 2 - r$ outside typed vertices.

At most one defective vertex. If $r = 0$, then $a = 0$ and $b = 2$, so at most two of the $2d - 2 \geq 12$ outside typed vertices lack degree d, and none has a helper. If $X = \{x\}$ and x has no cycle neighbour, then $b = 0$ and x must be adjacent to all $2d - 3 > d$ outside typed vertices. If x has one cycle neighbour, then $a = b = 1$ and x is adjacent to at least $2d - 4$ outside typed vertices besides its cycle neighbour, so $\deg x \geq 2d - 3 > d$.

Two defective vertices. Now $a = 2$, $b = 0$: every outside vertex has degree exactly d, $X = \{x_1, x_2\}$ with $x_1 \sim c_p$ and $x_2 \sim c_q$ as their only cycle neighbours, $\sum_i n_i = 2d + 1$ and $\sum_i \varepsilon_i = 2$. The vertex x_1 helps only vertices of $V_{p+2} \cup V_{p+3}$, and x_2 only vertices of $V_{q+2} \cup V_{q+3}$; a type with no possible helper is a singleton. For an outside $v \in V_i$ let $\xi(v) = |N(v) \cap X|$; then

$$|N(v) \cap W_i| = d - \xi(v). \tag{12}$$

Rule (T): if some $x \in X$ is adjacent to $u \in V_i$ and $w \in V_{i+1}$, then $u \not\sim w$. Up to rotation and reflection, $q - p \in \{0, 1, 2\}$.

Case $q = p$. Only V_{p+2}, V_{p+3} can be helped, so $n_{p-1} = n_p = n_{p+1} = 1$, and (11) at $c_{p\pm1}$ gives $n_{p+2} = n_{p+3} = d - 1$. Let A°, B° be the outside vertices of V_{p+2}, V_{p+3} ($d - 2$ each). By (12) an outside vertex of V_{p+2} misses exactly $\xi \leq 2$ vertices of V_{p+3}, and symmetrically. If x_k had neighbours on both sides, Rule (T) would bound its typed outside neighbours by four, while $\deg x_k = d$ forces at least $d - 2 \geq 5$. So x_1 covers A° and x_2 covers B°, say, and $d = \deg x_1 \leq 1 + (d - 2) + [x_1 \sim x_2]$ forces $x_1 \sim x_2$, a triangle with c_p.

Case $q = p + 1$. Here $n_p = n_{p+1} = 1$ and $\varepsilon_p = \varepsilon_{p+1} = 1$, so $n_{p+2} + n_{p+3} + n_{p+4} = 2d - 1$. Inequality (11) at $c_p, c_{p+1}, c_{p+2}, c_{p+3}$ gives $n_{p+4} \leq d - 2$, $n_{p+2} \leq d - 2$, $n_{p+3} \leq d - 1$ and $n_{p+2} + n_{p+4} \leq d$; with the sum this forces $n_{p+3} = d - 1$, $n_{p+2} + n_{p+4} = d$, and both $n_{p+2}, n_{p+4} \geq 2$. An outside vertex of V_{p+2} has x_1 as its only possible neighbour in X, so $\xi = 1$ and by (12) it misses exactly one vertex of V_{p+3}. As all outside vertices of V_{p+2} are adjacent to x_1, Rule (T) lets x_1 have at most one neighbour among the $d - 2 \geq 5$ outside vertices of V_{p+3}; likewise x_2. Those vertices cannot all be helped.

Case $q = p + 2$. Now $n_{p+1} = 1$ and $\varepsilon_p = \varepsilon_{p+2} = 1$. Write $\alpha, \beta, \gamma, \delta$ for $n_p, n_{p+2}, n_{p+3}, n_{p+4}$, so $\alpha + \beta + \gamma + \delta = 2d$, and (11) gives $\delta \leq d - 2$, $\alpha + \beta \leq d$, $\gamma \leq d - 2$, $\beta + \delta \leq d$, $\gamma + \alpha \leq d$; hence $\gamma, \delta \geq 2$. Outside vertices of V_{p+3} (respectively V_{p+4}) have x_1 (respectively x_2) as their only neighbour in X, so each has $d - 1$ neighbours in $V_{p+2} \cup V_{p+4}$ (respectively $V_{p+3} \cup V_p$). If $\beta = 1$ this forces $\delta = d - 2$ and then $\alpha + \gamma = d + 1$, a contradiction. If $\beta \geq 2$, Rule (T) at x_1 separates the outside vertices of V_{p+2} and V_{p+3}, so $\delta = d - 2$. If moreover $\alpha = 1$, then $\gamma = d - 2$ and $\beta = 3$, contradicting $\beta + \delta \leq d$. If $\alpha \geq 2$, Rule (T) at x_2 gives $\gamma = d - 2$, hence $\alpha = \beta = 2$, and the single outside vertex of V_{p+2} has degree at most $|\{c_{p+1}, c_{p+3}\}| + |X| = 4 < d$.

Every case is contradictory, which proves Theorem 9. □

An independent reading re-derived every case, including the reduction of the 25 ordered pairs (p, q) to the three cases above, and found no gap. A satisfiability search at $d = 7$, archived as EXP-004 [11], corroborates the statement; the universal claim rests on the proof.

The rejected complement shortcut

Although Q_d attains the triangle-free maximum, its complement fails the connectivity required above. Indeed,

$$N_{Q_d}(A_0) = N_{Q_d}(A_1) = \{v, B_2, \dots, B_d\}.$$

This gives a $K_{2,d}$ on $d+2$ vertices. Deleting the other $d+1$ vertices disconnects the complement, so $\kappa(\overline{Q_d}) \le d+1$.

In fact equality holds. A cut of size at most d would leave a complete bipartite subgraph of Q_d on at least $d+3$ vertices. Without v, its order is at most $d+1$ by the crown argument. With v in one side, the other side is contained in $\{A_0, A_1, B_0, B_1\}$. If it uses both original classes, no old vertex can join v in its side, and total order is at most five. If it uses only one original class, it has size at most two, and the degree bound at an opposite vertex bounds the side containing v by d. Its total order is at most $d+2$. Thus

$$\kappa(\overline{Q_d}) = d+1.$$

The shortcut fails because extremality under a degree bound does not supply the needed complement connectivity. Theorem 9 shows that the failure is forced: every fixed-order maximizer has two vertices with a common d-neighbourhood, as A_0, A_1 do in Q_d.

6 Exact checks, scope and remaining questions

The universal assertions follow from the proofs above. EXP-003 [10] additionally contains deterministic exact verification for $d = 7, \dots, 30$. For each degree, it saves both full edge lists, explicit matching witnesses, degrees, complement data and the rejected cut. A standard-library implementation checks all 48 graphs and performs direct flow checks on 24 graphs with $d \le 18$. It also checks 287,564 positive five-type tuples with a designated singleton; no equality candidate survives.

A separate implementation reconstructs all saved graphs in NetworkX 3.4.1 without importing the CAOS graph routines. It independently checks maximum matching, clique number, all 48 complement connectivities and all 24 negative cuts. It enumerates all 17,040 positive five-part compositions for $d = 7, \dots, 10$, with no singleton-containing equality survivor. The all-large case is already excluded symbolically. Both implementations detect a deliberately inserted triangle and verify the degree-six boundary control. Source and certificate hashes are persisted with the results. A regression test reproduces the main certificate.

A separate reading of the full proof checked the odd-degree cases, the complete singleton case split, the matching-budget reduction and both connectivity arguments. The proof has not been formalized in a proof assistant. Finite checks cannot rule out a gap in an all-parameter argument, and bounded literature searching cannot guarantee priority.

Theorem 1 settles the next-matching slice for every $d \ge 7$, and Theorem 8 the Bougard–Joret value at $(2d+3, 2, d+2)$. Neither settles the remaining intermediate matching levels or classifies all extremizers; Theorem 9 gives only the shared-neighbourhood property. The general revised Bougard–Joret regimes, including the second regime $n > k\alpha$, remain open. The earlier first interior shell result [9] is a separate theorem.

Data availability. The proof, the review of the cited sources, the certificate generators, the independent re-implementation and the full edge lists are publicly archived in the repository [10]. The identifier EXP-003 denotes the archived computation in the folder `problems/combinatorics/bougard-joret/experiments/EXP-003-triangle-free-next-matching/`. The sibling folder `EXP-004-bracket-endpoint/` holds EXP-004, the proof and satisfiability record for Theorem 9.

Declaration of generative AI use. Generative AI tools were used for literature triage, derivation exploration, code drafting and internal review of the proofs and the text. The author reviewed and verified all content and takes full responsibility for the work.

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